

Lesson 8: Demand theory part



Utility max vs Expenditure min

The twin with each other

$$P \gg 0, W > 0, V > V(0)$$

$$E(P, V(P, W)) = W \quad \text{and} \quad V(P, E(P, V)) = V$$

The first relation says the cheapest to obtain the highest utility to V can get

with wealth W is going to cost ex. W .

Otherwise, I have failed to solve ^{one of} two prob. correctly.

Important given prices, the two func. are inverse to each other. If I know one,

then I can infer to the other

For similar reasons, the solutions to the two prob. coincide in the sense that

$$H(P, V) = X(P, E(P, V)) \quad \text{and} \quad X(P, U) = H(P, V(P, W))$$

Hicksian demand sol. exp min prob.

The first equation also describes with hicksian

is sometimes called compensated demand.

When prices change, the consumer is compensated s.t.

his wealth is just enough achieve the same utility as before
↳ a.k.a hicksian ^{wealth} compensation.

Compare to the Slutsky we. compen. which allows the consumer to buy same bundle as before.

Assume the solution to the exp. min problem is unique. It's a demand function rather than a demand correspondence.

Hicksian demand also satisfies the law of compensated law of demand

$$(P' - P)(H(P', u) - H(P, u)) \leq 0$$

Easy proof. by definition of hicksian demand

$$P' \cdot H(P', u) \leq P' \cdot H(P, u)$$

$$\Leftrightarrow P'(H(P', u) - H(P, u)) \leq 0$$

$$P \cdot H(P, v) \leq P \cdot H(P', v) \Leftrightarrow \\ P \cdot (H(P', v) - H(P, v)) \geq 0$$

And the compensated law of demand follows.

To further explore the links between two versions

of the compensated law of demand. Start with

shephard's lemma.

$$H_i(P, v) = \frac{\partial E(P, v)}{\partial P_i}$$

Proof by en. Theron.

Try to prove it, Claude.

Now, consider an own price change

$$\frac{\partial H_i(P, v)}{\partial P_i} = \frac{\partial^2 E(P, v)}{\partial P_i^2} < 0 \text{ since exp. is concave in } P$$

Hicksian demand $\downarrow$ in its own price.

Consider next how demand for good i on price of Good j .

$$\begin{aligned}\frac{H_i(P, U)}{\partial P_j} &= \frac{\partial X_i(P, E(P, U))}{\partial P_j} \\ &= \frac{\partial X_i(P, E(P, U))}{\partial P_j} + \frac{\partial X_i(P, w)}{\partial w} \frac{\partial E(P, U)}{\partial P_j} \\ &= \frac{\partial X_i(P, w)}{\partial P_j} + \frac{\partial X_i(P, w)}{\partial w} H_i(P, U)\end{aligned}$$

$$\frac{\partial H_i(P, U)}{\partial P_j} = \frac{\partial X_i(P, w)}{\partial P_j} + \frac{\partial X_i(P, w)}{\partial w} X_i(P, w)$$

Same expression for the Slutsky substitution matrix.

So now we know that the terms in the Slutsky substitution matrix

as of the form $\frac{\partial^2 H_i(P, U)}{\partial P_i^2}$

or $\frac{\partial^2 E(P, U)}{\partial P_i \partial P_j}$ is just the Hessian of $E(P, U)$

Since $E(P, U)$ is concave. This is a negative semi-definite matrix.

The matrix is also symmetric. This is new and not something

got out of WARP. However, SARP can be shown to give symmetry.

Welfare evaluations:

What happens to the consumer's well-being

if P or w changes? Focus on P .

Qualitative easy if $\bar{u}$ is known. Since all we need to ask is whether utility increases or decreases. The indirect utility function, $v(P, w)$, is ideal for this purpose.

But there's no quantitative interpretation.

Given $\succeq$, $V(x)$ is not unique so $V(p, w)$ is not unique either.

We develop some measures that have quantitative meaning.

We develop a particular kind of indirect utility func.

Later, the welfare to Hicksian demand and contrast to Marshallian demand.

Start with $\succeq$. Pick any $V(x)$ that pre. $\succeq$.
Derive $V(p, w)$ and $E(p, V)$. And now, we construct

another, the money metric indirect utility func.

Fix some arbitrary price vector $\bar{p} \gg 0$ and calculate

$$V_m(p, w | \bar{p}) = E(\bar{p}, V(p, w))$$

This is an indirect utility function it maintains the same order - the transformation of the right is monotone.

Additionally it's the same regardless of which $V(x)$ represents $\bar{z}$ that we started with

$V_m(p, u(\bar{p}))$ is just the smallest amount of money needed to

But the best bundles under (p, w) But this is determined by $\bar{z}$ and not by $u(x)$